**In the Name of God, the Creator of All Crafts**

# Management Information Systems (MIS) Project

**Instructor: Dr. Mossadegh**

**Teaching Assistants:** Hadith Gorjali, Mozhdeh Shakerian, Faridoddin Oloomi

## 1 Comprehensive Overview of the Copper Extraction Process

Copper extraction comprises an interconnected sequence of processing stages designed to liberate, extract, and refine elemental copper from raw ore deposits. The lifecycle initiates with the excavation of raw material in open-pit or underground mines and systematically proceeds through crushing, concentration, smelting, and refining, yielding high-purity industrial copper cathodes.

- **Drilling and Blasting:** Ground fragmentation operations break rigid in-situ rock into transportable ore blocks.
- **Haulage & Comminution:** Excavated materials are loaded and conveyed to mineral processing units, where primary/secondary crushing and grinding reduce rock dimensions to liberate copper-bearing minerals from gangue.
- **Concentration (Froth Flotation):** Finely ground slurry undergoes froth flotation to physically segregate hydrophobic copper sulfides from tailings, producing high-grade copper concentrate.
- **Smelting and Electrorefining:** The concentrate is thermally processed in pyrometallurgical furnaces to extract anode copper, which subsequently undergoes electrorefining to produce premium electrolytic copper cathodes (99.99% purity).
- **Commercial Products:** Refined copper is cast into commercial forms, including cathodes, ingots, and continuous rods distributed across international manufacturing industries.

### 1.1 Industrial Lifecycle Sub-processes

**1. Drilling & Blasting**

**Input:** Copper-bearing ore bodies (varying grades).
**Output:** Fragmented ore suitable for haulage to the comminution plant.
**Process Steps:**

1. **Geological Survey:** Delineating prospective drilling coordinates via initial analytical data.
2. **Drilling:** Boring blast holes at designated coordinates to predetermined depths.
3. **Explosive Loading:** Charging explosive agents (dynamite/ANFO) into blastholes.
4. **Blasting:** Controlled detonation to fracture bulky rock masses into manageable sizes.
5. **Debris Gathering:** Mucking and staging fragmented ore for loading operations.

### 2. Raw Material Transportation (Haulage)

**Input:** Fragmented blasted ore from the open pit.
**Output:** Sized feed staged at the primary crusher intake dump hopper.
**Process Steps:**

1. **Loading:** Loading muck piles into haul trucks or transfer conveyor belts using shovels/excavators.
2. **Hauling:** Haulage cycle toward the crushing facility along engineered haul roads.
3. **Unloading:** Dumping ore directly into the primary crusher bin or run-of-mine (ROM) stockpile.

### 3. Crushing & Screening

**Input:** Raw bulk copper ore (large dimensions).
**Output:** Calibrated crushed rock sized for downstream milling/grinding circuits.
**Process Steps:**

1. **Primary Crushing:** Primary size reduction using heavy-duty Jaw Crushers.
2. **Secondary Crushing:** Secondary size reduction utilizing hydraulic Cone Crushers.
3. **Screening:** Vibrating screen classification separating fines from oversized particles.
4. **Mill Feed Conveyance:** Classified fine material flows to the grinding circuit; oversize fractions return via closed loop to the secondary crushers.

### 4. Milling & Leaching

**Input:** Crushed ore of calibrated dimensions.
**Output:** Micronized mineral slurry containing dissolved copper ions.
**Process Steps:**

1. **Ball Mill Grinding:** Micronization of ore particles inside tumbling ball mills.
2. **Slurry Preparation:** Blending pulverized mineral fines with water to generate stable pulp.
3. **Chemical Reagent Dosing:** Injecting conditioning chemicals to prepare pulp for leaching.
4. **Acid Leaching:** Aqueous dissolution of copper oxides/silicates using dilute sulfuric acid.
5. **Slurry Pumping:** Pressurized pumping of pulp to subsequent concentration/separation units.

### 5. Concentration & Flotation

**Input:** Mineral slurry containing liberated copper compounds.
**Output:** High-purity, dewatered copper concentrate cake.
**Process Steps:**

1. **Reagent Conditioning:** Introducing collectors and frothers rendering copper minerals hydrophobic.
2. **Froth Formation:** Sparging compressed air through flotation cells, attaching mineral particles to rising bubbles.
3. **Froth Skimming & Separation:** Laundering the mineral-laden froth; discharging gangue tailings.
4. **Concentrate Dewatering:** Thickening, filtration, and drying to remove excess process water.
5. **Transfer to Smelter:** Dispatching dry concentrate cake to pyrometallurgical facilities.

### 6. Smelting & Refining

**Input:** Copper concentrate cake.
**Output:** Commercial Electrolytic Copper Cathodes (99.99% Cu purity).
**Process Steps:**

1. **Furnace Smelting:** Thermal reduction at high temperatures to eliminate iron and gangue as slag.
2. **Pyrometallurgical Converting:** Oxidizing molten matte to eliminate sulfur, casting copper anodes ($98.5\% - 99.5\%$).
3. **Electrorefining:** Electrolytic dissolution of anodes onto starter sheets in aqueous $CuSO_4$, yielding 99.99% pure copper cathodes.
4. **Bundling & Warehousing:** Strapping cathodes into standardized strapped bundles for shipment.

### 7. Warehouse & Inventory Management

**Input:** Raw materials, consumables, work-in-progress (WIP), and finished cathodes.
**Output:** Optimized safety stock levels and traceably managed inventory.
**Process Steps:**

1. **Finished Goods Warehousing:** Shelving, indexing, and storing concentrates and cathodes.
2. **Reagent Management:** Procuring and staging process acids, flotation collectors, and activated carbon.
3. **Stock Control:** Real-time reconciliation preventing stockouts or inventory holding excesses.

## 8. Quality Control & Copper Grade Monitoring

**Input:** Representative specimens from ROM ore, intermediary slurry, concentrate, and refined metal.
**Output:** Analytical laboratory assays documenting chemical composition and grade recovery.
**Process Steps:**

1. **Systematic Sampling:** Automated and manual pulp/core sampling across processing nodes.
2. **Spectrometric Assaying:** Wet chemical and X-ray fluorescence (XRF) grade determination.
3. **Process Tuning:** Reconciling actual grades against targets and issuing setpoint corrective feedback.

## 9. Downtime & Breakdown Management

**Input:** Automated SCADA anomaly logs and maintenance work request orders.
**Output:** Maximized Overall Equipment Effectiveness (OEE) and reduced MTTR.
**Process Steps:**

1. **Mechanical Fault Identification:** Diagnosing failures in crushers, mills, and smelting furnaces.
2. **Process Anomaly Diagnostics:** Root-cause analysis of flotation yield declines or head-grade shifts.
3. **Logistics Bottleneck Control:** Resolving haul truck dispatch jams and fuel/feed delivery delays.
4. **Preventive/Corrective Action:** Executing standardized maintenance interventions and operational recovery.

# 2 Project Roadmap: Phases 0 through 3

**Context:** As an appointed Management Information Systems (MIS) project team member within a primary copper manufacturing enterprise over a three-month operational cycle, you are tasked with executing the following four sequential phases:

## 2.1 Phase 0: Process Architecture & Algorithmic BPMN Definition

### 2.1.1 1. Core Scope and Deliverables

- **Lifecycle Examination:** Analyze the nine core operational sub-processes governing copper extraction.
- **Selection of Five Operations:** Select **five (5)** critical production-related processes to model in detail (e.g., Drilling, Haulage, Primary Crushing, Inventory Control, Quality Sampling).
- **Textual Model Formulation:** Specify each selected process rigorously using standard BPMN entities: actors (swimlanes), start triggers, sequential activities, decision gateways, and termination events.

### 2.1.2 2. Reference Blueprint (Example: Drilling Process)

The following standard blueprint exemplifies the expected level of analytical depth:

**A. Swimlanes (Organizational Roles):**

1. **Geology Department**
2. **Drilling Engineering Team**
3. **Drilling Rig Operator**
4. **Safety & Environmental Team**

**B. Process Flow Logic:**

1. **Process Initiation (Start Event):**
   - *Input Data:* Geological exploration report and mine bench drilling patterns.
   - *Start Trigger:* Formal issuance and reception of the drilling authorization permit.
2. **Drilling Preparation Stage (Geology & Drilling Engineering):**
   - Review geological structure; stake specific blasthole coordinates.
   - Calculate hole diameter, burden, spacing, and target drilling depth.
   - Assess seismic, terrain, and environmental hazard constraints.
   - **Exclusive Gateway [Approval Required?]:**
     - **Yes:** Forward operational clearance to the field rig team.
     - **No:** Route back for engineering revision and geotechnical recalibration.
3. **Rig Setup Stage (Drilling Operator):**
   - Mobilize and level drilling rig over surveyed hole collar.
   - Adjust mast inclination, drill string angle, and initial bit depth.
   - Complete pre-start mechanical checklists and inspect emergency kill switches.
   - **Exclusive Gateway [Equipment Inspection]:**
     - **Pass (Ready):** Advance immediately to active rotary/percussion drilling.
     - **Fail (Defective):** Tag out rig, report alert, and trigger field mechanical maintenance.
4. **Drilling Execution Stage (Drilling Operator):**
   - Initiate drilling with appropriate rotation speed, feed pressure, and flushing air.

- Continuously log penetration rate and depth telemetry.
- Extract and inspect drill chip cuttings at planned intervals.
- Commit operational log data to the mine field terminal.
- **Exclusive Gateway [Target Depth Reached?]:**
  - **Yes:** Terminate drilling string rotation; proceed to hole clearing.
  - **No (or Geological Deviation):** Adjust torque/thrust parameters or replace worn drill bit.

5. **Completion & Technical Evaluation (Engineering & Safety):**
   - Log finished hole depth, azimuth, and collar coordinates.
   - Transmit verified hole log records into the central Mine Dispatch System.
   - Perform environmental and bench stability assessment.
   - Publish final signed Drilling Completion Report.
6. **Process Termination (End Event):**
   - Blast hole array accepted; zone cleared and transferred to the Blasting & Haulage squad.

## 2.2 Phase 1: Visual Modeling in Visual Paradigm (BPMN 2.0)

- **Visual Paradigm Diagram Generation:** Convert the Phase 0 structured workflows into graphical BPMN 2.0 diagrams inside Visual Paradigm.
- **Notation Fidelity:** Strictly deploy standard visual elements:
  - **Pools & Swimlanes:** Partitioning distinct functional units and user roles.
  - **Events:** Clearly differentiated Start, Intermediate, and End circular events.
  - **Activities & Tasks:** Standard rounded rectangles categorized as User, Service, or Manual tasks.
  - **Gateways:** Diamond shapes for exclusive (XOR), parallel (AND), or inclusive (OR) branching.
  - **Sequence Flows:** Unbroken arrows governing strict chronological message/control flow.
- **Metadata Documentation:** Integrate complete task descriptions, inputs, outputs, and decision condition expressions within the diagram properties.

## 2.3 Phase 2: Relational Data Modeling and Enterprise Database Population

- Design normalized Relational Schemas (ERD / Conceptual-Logical Data Models) supporting all five modeled business processes.
- Generate realistic production datasets representing factory telemetry, equipment logs, assay results, inventory movements, and personnel records.
- Implement and populate database tables via structured SQL schemas.

## 2.4 Phase 3: Executive Business Intelligence (BI) Dashboards

- Design interactive, multi-tier executive dashboards (e.g., using Power BI / Tableau / Metabase).
- Visualize critical performance metrics: Plant Overall Equipment Effectiveness (OEE), Copper Grade Yields vs. Targets, Inventory Turnovers, and Equipment Downtime/MTBF/MTTR analytics.